# Hoashalarajh Rajendran

*Electrical Engineer*

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## Professional Summary


Innovative Electrical Engineer holding a B.Sc. (Hons) in Electrical Engineering, from the University of Moratuwa. Specializes in power systems. Proven ability to model power system components, conduct load flow and fault analysis studies, recommend power cable sizing and specifications for suitable applications by conducting thermal and electrical analysis, and implement advanced decision-making models utilizing AI. Seeks to leverage strong analytical capabilities within a vibrant engineering environment to solve complex engineering problems.


## Core Technical & Industrial Skills

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|---|---|
| Power & Industry Software | DIgSILENT PowerFactory, ETAP, Siemens PSS®E, MATLAB & Simulink, AutoCAD, Siemens Tecnomatix Plant Simulation, Siemens PLC (S7-1200 TIA Portal), SCADA/HMI systems |
| Competencies | Power System Modeling & Analysis, Protective Relay Coordination, Contingency analysis, Preventive Maintenance, High-Voltage Engineering, IEC Standards Testing, Optimization, Discrete Event Modeling & Simulation |
| Programming | MATLAB, Python |
| Advanced Tools | Electrical Engineering Drawings & Documentation, Cable Design & Specification, Power System Protection & Fault Analysis, Digital Twins, Key Performance Indicators (KPIs) |

## Professional Industrial Experience

Jun 2026–Present **Assistant Electrical Engineer**, *10 MW Biomass Power Plant, Tokyo Cement Company (Lanka) PLC*, Trincomalee, Sri Lanka

- Executed preventive and corrective maintenance on the 10 MW biomass power plant's core electrical infrastructure, including synchronous generators, excitation systems, and prime mover controls.
- Monitored and managed power generation, overseeing grid synchronization, high-voltage switchgear operations, power transformers, and load dispatch for facility-wide distribution.
- Diagnosed and maintained power plant DCS/SCADA frameworks, performing field instrument calibration and troubleshooting control loops for boiler and turbine operations.
- Updated critical electrical documentation, including generation maintenance logs and single-line diagrams (SLDs).

May 2021–Jun 2021 **Electrical Engineering Trainee**, *Ceylon Electricity Board (CEB) - Upper Kothmale Hydropower Station*

- Observed daily plant operations, including unit start-up/shut-down procedures and grid synchronization via SCADA systems.
- Assisted in the monitoring and analysis of synchronous generators, excitation systems, and step-up transformers.
- Reviewed electrical single-line diagrams, control schematics, and operational manuals for hydropower generation equipment.
- Supported engineers during routine maintenance, condition monitoring, and safety inspections of switchgear and auxiliary systems.
- Assisted with compiling daily generation logs and technical documentation for preventive maintenance schedules.

Apr 2021–May 2021 **Electrical Engineering Trainee**, *Lakdhanavi (Pvt) Ltd - Yugadanavi 300 MW Combined Cycle Power Plant*

- Monitored and analyzed real-time electrical telemetry data for a 300 MW combined cycle facility using Distributed Control Systems (DCS).
- Conducted routine field inspections and condition monitoring on heavy industrial electrical machines, including generators, large induction motors, and high-voltage switchgear.
- Collaborated with plant engineers to execute preventive maintenance schedules and investigate operational anomalies in electrical subsystems.
- Compiled technical performance reports and updated critical system documentation, including maintenance logs and single-line diagrams.

Jan 2021–Apr 2021 **Electrical Engineering Trainee**, *LTL Transformers (Pvt) Ltd*

- Monitored the end-to-end distribution transformer manufacturing lifecycle, including core building, coil winding, active part assembly, and vacuum oil-filling processes.
- Conducted routine and type testing on distribution transformers (winding resistance, insulation resistance, and dielectric tests) to verify compliance with international IEC standards.
- Evaluated shop-floor production workflows and quality control checkpoints, assisting with defect tracking and process optimization.
- Reviewed engineering design blueprints, CAD models, and technical specifications related to transformer production.

2022–2024 **Graduate Research Assistant**, *Electrical Engineering*, University of Moratuwa, Sri Lanka

- Designed context-aware architectures and machine learning frameworks for hardware implementations, for human-friendly conversational agent systems.

2025 **Graduate Research Assistant**, *Faculty of Sustainable Design Engineering*, University of Prince Edward Island, Canada

- Focused on systems modeling, simulation, and long-horizon decision-making algorithms, applying optimization techniques to dynamic, uncertain environments for mobile robots.

## Education

2022–2024 **M.Sc. in Electrical Engineering**, *University of Moratuwa*
*Thesis:* Situation-aware proactive services by observing user behaviors for domestic service agents
*Supervisor:* Prof. Buddhika Jayasekara, Department of Electrical Engineering, UoM
*Coursework:* Artificial Intelligence Applications - *Grade:* A
*Specialization:* Intelligent systems & Sustainability

2017–2022 **B.Sc. Engineering (Hons) in Electrical Engineering**, *University of Moratuwa*
*Academic Standing:* Second Class Upper Division
*FYP:* Digital Twin of an Intelligent Production Line with Adaptive Resource Handling
*Skills:* Data Collection, Cleaning, Analysis, Probabilistic Modeling, Simulation, Computing & Comparing Key Performance Indicators (KPIs)

Additional Coursework Operations Research, Renewable Energy, Modeling and simulation, Image Processing, Data Analysis

2017 **Diploma in AutoCAD**, *Esoft Metro Campus*
*Academic Standing:* Distinction

Skills Engineering drawing, 2D Drafting, 3D Modeling and Rendering, Mechanical and Electrical Design

## Key Engineering & Systems Projects

2021–2022 **Digital Twin of Intelligent Production Line**, *Final Year Project*, Grade:**A**

*Supervisor:* Prof. Buddhika Jayasekara, Department of Electrical Engineering, UoM

- Constructed a complete digital representation of an electrical transformer production line.
- Analyzed bottlenecks and estimated KPIs such as Throughput, Cycle Time, and Overall Equipment Efficiency (OEE).
- Designed and integrated a fuzzy-logic decision-support system, increasing simulated system throughput by 12.5%.

2025 **Socially-aware Autonomous Navigation Architecture**, Canada

- Architected long-horizon decision-making systems for autonomous mobile agents operating in highly uncertain, dynamic crowd environments.
- Modeled crowd dynamics (density, flow, proximity) utilizing Partially Observable Markov Decision Processes (POMDP) to shift from reactive navigation to strategic, socially-compliant movement.
- Implemented belief-space planning algorithms using Monte Carlo Tree Search (MCTS) to optimize system decision-making under uncertainty.

2025 **Smart Face Detection & Emotion Estimation**, *ROS2, Python, OpenCV*

- Engineered a modular ROS2 (Humble) Python ecosystem bridging live video feeds with computer vision.
- Deployed a publisher-subscriber architecture to broadcast video frames, executing real-time human frontal face detection and instantaneous emotion intensity estimation.

2022–2024 **Context-Aware Conversational Agent for Human-friendly Robots**, Sri Lanka

- Developed hierarchical state transition networks for complex conversational task management.
- Engineered proactive systems to analyze user behavior patterns by integrating multi-modal data streams (voice, vision, text).
- Built a real-time interaction manager capable of dynamically switching states based on user behavior, improving continuous and supportive human-agent interaction.

## Professional Memberships

IESL Associate Member (AMIE) of the Institution of Engineers, Sri Lanka (IESL)

IEEE Graduate Student Member, Institute of Electrical and Electronics Engineers

IEEE CIS Member, IEEE Computational Intelligence Society

IEEE SMC Member, IEEE Systems, Man, and Cybernetics Society

IEEE YP Member, IEEE Young Professionals

## Awards & Publications

2023 Best Paper Award - Robotics and Intelligent Automated Systems Track (IEEE MERCon 2023)

2023 Published 5 peer-reviewed papers in IEEE conferences (R-10 HTC, ICIIS, SMC, MERCon)

2020 Dean's List, Faculty of Engineering, University of Moratuwa

## References

**Prof. A.G.B.P. Jayasekara, Ph.D. (Saga University, Japan), B.Sc. Eng, M.Sc (University of Moratuwa)**
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